Practice Problems

July 2026

1 Numbers

1. Write a script that assigns the floating-point literal 175000.0 to the variable `num` using exponential notation, and then prints `num` in the interactive window.

2 Script

1. Initialise the string ”lalalalal admamda ”
2. access the element number
3. concatenate it with
4. display the concatenated string
5. Create a string and print its length using the len() function.
6. what happens when you try to access the element no.
7. try different combinatioin of the access command to a string entry to see what results do they yield. for example ”flavor[:5]”
8. try negative indices flavor[-9:-6]
9. Use concatenation to create a new string from an old string by changing only a few letters.
10. Print the string ”zing” by using slice notation on the string ”bazinga” to specify the correct range of characters.
11. converting string to all lower case letters `.lower()`
12. Try `name = "Jean-luc Picard"`
    >>> `name.lower()`.
13. `.upper()` method.
14. Try `len()` function

15. Use IDLE to Discover Additional String Methods
    `>>> starship = "Enterprise"`

16. Try
    ```
    >>>  num = "2"
    >>>num + num
    ```

17. Strings can be "multiplied" by a number as long as that number is an integer, or whole number. Try:
    ```
    >>> num = "12"
    >>> num * 3
    >>>  "12" * "3"
    >>> "3" + 3
    ```

18. Try `int("12")`

19. `str()` can even handle arithmetic expressions: Try
    ```
    >>> total_pancakes = 10
    >>> pancakes_eaten = 5
    >>> "Only " + str(total_pancakes - pancakes_eaten) + " pancakes
    left."
    ```

20. Find a String in a String:
    ```
    >>> phrase = "the surprise is in here somewhere"
    >>> phrase.find("surprise")
    ```

21. check with a different string that's not present and what is the result.

22. Try `>>> "My number is 555-555-5555".find(5)`
    and `>>> "My number is 555-555-5555".find("5")`

23. Just like `.find()`, you tack `.replace()` on to the end of a variable or string literal.
    ```
    >>> my_story = "I'm telling you the truth; nothing but the truth!"
    >>> my_story.replace("the truth", "lies")
    ```

24. strings are immutable objects right!!!

3 Input

1. To see how `input()` works, save and run the following script:
    ```
    prompt = "Hey, what's up?  "
    user_input = input(prompt)
    print("You said:", user_input)
    ```

2. Write a script that takes input from the user and displays that input back.

3. Write a script that takes input from the user and displays the number of characters inputted.

4. Write a script that gets two numbers from the user using the `input()` function twice, multiplies the numbers together, and displays the result. If the user enters 2 and 4, your program should print the following text: `The product of 2 and 4 is 8.0.`

5. Write a script called `exponent.py` that receives two numbers from the user and displays the first number raised to the power of the second number.

6. Write a script that asks the user to input two numbers by using the `input()` function twice, then display whether or not the difference between those two number is an integer.

7. Print Numbers in Style: